Introduction to Web

Agenda

1. Internet and WWW
2. Web server and HTTP
3. HTTP requests and responses

1. Internet and WWW

- What is internet
  - It is the massive network of networks.
  - It builds physical infrastructure for running multiple services.
  - Technically, no one runs the Internet. No one acts as administrator of the internet. Everyone can just be the part of the internet.
  - There can be milions of hosts on the internet
- History of internet
  - Origin of internet lies in ARPANet
  - ARPANet : It is the project undertaken by Department of Defense as a way to protect government communications systems in the event of a military strike.
  - In late 1980's it grew into massive network where many universities, research institutes, government networks became part of it

- Working of internet
  - It allows many different computers to connect and talk to each other.
  - This is possible because of a set of standards, known as protocols, that govern the transmission of data over the network: TCP/IP (Transmission Control Protocol/Internet Protocol)
  - TCP/IP is normally considered to be a 4 layer system. The 4 layers are as follows:
    Application layer
    Transport(TCP) layer
    IP layer
    Hardware layer

1. Internet and WWW

- World Wide Web
  - *World Wide Web* is **not** synonymous with *the Internet*. It is a service running on the internet.
  - The Web, or World Wide Web, is basically a system of interlinked hypertext documents.
- URL
  - Universal Resource Locator, is the address of a resource on the Internet.
  - URL starts with protocol followed by domain name or IP address and optional port number which is followed by location of the resource.

1. Internet and WWW

- Domain name systems
  - Domain name systems help to translate user friendly domain names into IP address which is used to identify machine on the network.
  - Domain name servers holds distributed databases of URLs mapped with IP addresses.

2. Web server and HTTP

- A web server is a software that stores web pages, processes client requests, and delivers web pages to the client.
- Examples of web server : IIS, Apache
- Difference between web server and web application
- Web server and browser communicate using HTTP protocol.
- HTTP is application level protocol which sits on the top of communication protocol i.e. TCP/IP.
- HTTP decides the standard the way client sends the request to web server and the way web server sends the response to the client
- HTTP methods

3. HTTP requests and responses

- HTTP requests
  - Every HTTP request contains three elements which are:- Request Line, Request Header, Body of Request(optional).
  - Start line: HTTP method followed by request URL followed by HTTP version
  - Request headers: multiple key-value pairs
  - Body: May not be required in every requests.

3. HTTP requests and responses

- HTTP Response
  - It is received from the server consists of:-
  - Status line:- HTTP version followed by response status code followed by status text
  - Response Header:- There can be one or more response header similar like request headers
  - Response Body:- The response body contains the resource demanded by the client